

# Chaithanya Naik

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## RESEARCH INTERESTS

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Quantum Computing, Deep Learning, Reinforcement Learning, Scalable systems and Automated Reasoning

## EDUCATION

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**University of Wisconsin-Madison** 2022-present  
*PhD student in Computer Sciences*  
Advisor: Prof. Swamit Tannu

**Indian Institute of technology Bombay** 2016-2021  
*M.Tech & B.Tech in Computer Science and Engineering*  
*Minor in Physics*  
*Minor in AI and Data Science*  
Advisors: Prof. Sai Vinjanampathy, Prof. Ashutosh Gupta, Prof. Amit Sethi

## PUBLICATIONS

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**Scaling Qubit Readout with Hardware Efficient Machine Learning Architectures.**  
Sativik Maurya, **Chaithanya Naik Mude**, William D. Oliver, Benjamin Lienhard, Swamit Tannu.  
*50th International Symposium on Computer Architecture (ISCA), 2023, Orlando, Florida, USA.*

## ON-GOING RESEARCH PROJECTS

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**Qubit Mapping and Routing with Formal Methods** *Fall 2023 - Ongoing*  
Advisor: Prof. Swamit Tannu *R&D Project, UW-Madison*

- Analyzing existing methods that employ heuristics & SAT/SMT solvers to decipher efficient DSL & specification
- Probing for relatively simpler abstraction that can solve for dynamically re-configurable qubit systems

**ML for Quantum Circuit Synthesis and Compilation** *Summer 2023 - Ongoing*  
Advisor: Prof. Swamit Tannu *R&D Project, UW-Madison*

- Designing ML based solutions to adapt for dynamically changing requirements with minimal modifications
- Leveraging the use of RL techniques for adaptability and LLMs for reasoning and encoding into a DSL

## RESEARCH EXPERIENCE

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**Effects of leakage in superconducting systems** *Summer 2023*  
Advisor: Prof. Swamit Tannu *R&D Project, UW-Madison*

- Experimented on IBM systems to analyze the impact of leakage errors on efficacy of QEC, mainly on surface codes

**Robust Quantum Optimal Control** *Spring 2021*  
Advisor: Prof. Sai Vinjanampathy *Master's Thesis Project, IIT Bombay*  
*Part of work presented at APS March meeting 2022*

- Designed encoder-decoder based Seq2Seq learning model to generate noise-resistant optimized control sequences
- Analyzed literature on Geodesic analysis, Deep Sequential models, and Optimal Quantum Control methods

**Controller Synthesis** *Spring 2021*  
Advisor: Prof. Ashutosh Gupta *R&D Project, IIT Bombay*

- Synthesised a controller for real-time system, based on data-driven RL and algorithmic SMT/SAT approaches
- Studied about determinization and minimization of timed automata for a specification given in duration calculus

**Interactive Image Segmentation** *Fall 2020*  
Advisor: Prof. Amit Sethi *Bachelor's Thesis Project, IIT Bombay*

- Automated the generation of markers required to segment desired object in image with minimal user intervention
- Modified SeedNet, a DQN approach over MSRA10k dataset to achieve IOU of around 60.5

## PROFESSIONAL EXPERIENCE

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### VMware Inc.

Avi NSX-T load-balancer | Mentor: Anurag Palsule

July 2021 - August 2022

Bangalore, India

- Enhanced the performance of load balancer to manage heavy workloads through kubernetes operator
- Worked on a open source project for a kubernetes operator, catering to needs for NSX-T and openshift platforms

### Amazon Development Centre India Pvt. Ltd

Auto Trouble Ticket Manager | Mentor: Archit Agrawal

May - July 2019

Hyderabad, India

- Designed **runtime compilation** module using **Java Compiler API** to implement dynamic code execution
- Developed a **generic framework** to plug-in their models for information retrieval and auto-suggesting resolution

## ACADEMIC PROJECTS

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- **Quantum Divide & Conquer** | Analyzed possible extensions to the paradigm | Quantum Algorithms *Spring 2023*
- **UnrealSynthesisEngine** | Proof Verifier for Unrealizability Logic | Program Synthesis *Fall 2022*
- **Gomoku RL Playing Agent** | Fundamentals of Intelligent agents *Fall 2020*
- **Depth Map Prediction From Single Image** | Computer Vision *Spring 2019*
- **Compiler for C-like language** | Implementation of Programming Languages *Spring 2019*
- **Fake News Detection by Crowdsourcing** | Database and Information Systems *Fall 2018*
- **Micro-architectural Attacks** | FLUSH+RELOAD, DRAMA Template | Computer Architecture *Fall 2018*
- **3D Modelling and Animation** | Computer Graphics *Fall 2018*

## AWARDS & ACHIEVEMENTS

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- Secured **2nd** position in HCL HACK IITK, the country's largest hackathon in **cybersecurity**, by IIT Kanpur *2022*
- Honourable Mention in **NSUCrypto-2021**, 8th International Olympiad in Cryptography *2021*
- Awarded **Gold Medal** in **Bosch's Route Optimization** challenge at the 8th Inter IIT Tech Meet *2019*
- Secured All India Rank **340** in IIT-JEE Advanced | All India Rank **90** in JEE Main Paper-II *2016*
- Amongst **Top 300** students in the country qualified for **INPhO** (for two straight years) and **INMO** *2015,14*
- Awarded **KVPY** Fellowship by IISc Bangalore | Recipient of **NTSE** Scholarship by NCERT *2014,12*

## PROFESSIONAL RESPONSIBILITIES

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### Teaching Assistant

- Data Science Programming I | Prof. Michael Doescher *Fall 2022*
- Quantum Computing and Information | Prof. Sai Vinjanampathy *Spring 2021*
- Python programming lab | Prof. Sai Vinjanampathy *Fall 2020*

### Mentor

Quantum Computing, Computer Vision: Maths and Physics Club

April - June 2020

- Mentored a group of 4 students for the project, Spoof-Resistant Facial Recognition using Deep Learning
- Mentored a group of 6 students for designing Rubik's Cube Solver using Reinforcement Learning
- Mentored a group of 6 students for creating star hopping guide & creating astronomy website using Django
- Helped a total of 12 students in pursuing their interests in Quantum Computing, Astronomy & Computer Vision

Department Academic Mentor | CSE Department, IIT Bombay

June 2020 - July 2021

Department General Secretary | CSE Department, IIT Bombay

April 2019 - July 2020

## TECHNICAL SKILLS

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**Programming & Tools** C/C++, Python, Bash, Go-Lang, Qiskit, Git, pennylane  
**Deep Learning** PyTorch, TensorFlow, Keras, TensorFlow Quantum

## MISCELLANEOUS

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- Part of the Inter IIT contingent securing **Runner's Up Position** at Inter IIT Tech Meet held at IIT Bombay *2018*
- Attended **Vijyoshi Science Camp** organized by Indian Institute of Science (**IISc**), Bengaluru, India *2015*